

27 Supersymmetric Gauge Theories

27.8 Another Approach: Gauge-Invariant Supersymmetry Transformations⁵

It is somewhat disturbing that the supersymmetry transformation rules discussed so far involve ordinary spacetime derivatives, not gauge-invariant derivatives. For instance, in a $U(1)$ gauge theory, the transformation of the component fields of a chiral scalar superfield is given by Eqs. (26.3.15)–(26.3.17) as

$$\begin{aligned}\delta\psi_L &= \sqrt{2}\partial_\mu\phi\gamma^\mu\alpha_R\phi + \sqrt{2}\mathcal{F}\alpha_L, \\ \delta\mathcal{F} &= \sqrt{2}(\bar{\alpha}_L\not{\partial}\psi_L), \\ \delta\phi &= \sqrt{2}(\bar{\alpha}_R\psi_L).\end{aligned}\tag{27.8.1}$$

One might have thought that in the transformation of a chiral superfield that carries $U(1)$ charge q , the ordinary spacetime derivatives in Eq. (27.8.1) should be replaced with gauge-covariant derivatives, given in terms of the $U(1)$ gauge field V_μ by

$$D_\mu = \partial_\mu - iqV_\mu.\tag{27.8.2}$$

With such gauge-invariant supersymmetry transformations of the chiral superfields, one would still attempt to formulate supersymmetry transformations of the gauge supermultiplet that involve only the physical and auxiliary fields V_μ , λ , and D :

$$\begin{aligned}\tilde{\delta}V_\mu &= (\bar{\alpha}\gamma_\mu\lambda), \\ \tilde{\delta}\lambda &= iD\gamma_5\alpha + \frac{1}{2}[\partial_\mu\psi, \gamma^\mu]\alpha, \\ \tilde{\delta}D &= i(\bar{\alpha}\gamma_5\not{\partial}\lambda),\end{aligned}\tag{27.8.3}$$

in which ordinary spacetime derivatives appear because the gauge superfield carries no $U(1)$ charge.

This doesn't work. The algebra of these transformations does not close: the commutator of two of the modified supersymmetry transformations is not a linear combination of bosonic symmetry transformations, such as spacetime translations and gauge transformations. It follows that it is not possible to construct a Lagrangian for the chiral and gauge superfields that would be invariant under these modified supersymmetry transformations, because if there were such a Lagrangian then it would have to be also invariant under the commutators of these transformations, so that these commutators would have to be bosonic symmetries of the Lagrangian.

In 1973 de Wit and Freedman [11] showed that the supersymmetry algebra could be made to close by modifying the supersymmetry transformation properties of the chiral superfields not only by changing ordinary derivatives to gauge-invariant derivatives, but by also adding an extra term in the transformation of the $\mathcal{F}$ -component, so that for $U(1)$

⁵This section lies somewhat out of the book's main line of development and may be omitted in a first reading.

gauge theories the modified supersymmetry transformation rules read

$$\begin{aligned}\tilde{\delta}\psi_L &= \sqrt{2} D_\mu \phi \gamma^\mu \alpha_R \phi + \sqrt{2} \mathcal{F} \alpha_L, \\ \tilde{\delta}\mathcal{F} &= \sqrt{2} (\bar{\alpha}_L \not{D} \psi_L) - 2iq\phi(\bar{\alpha}_L \lambda_R), \\ \tilde{\delta}\phi &= \sqrt{2} (\bar{\alpha}_R \psi_L).\end{aligned}\tag{27.8.4}$$

With this change, they were also able to construct a Lagrangian that is invariant under the transformations (27.8.3)–(27.8.4) and that turned out to be just the one we have found in Sections 27.1 and 27.2.

There is nothing wrong with continuing to use the conventional transformation rules (27.8.1), so we do not need the de Wit–Freedman formalism to deal with supersymmetric gauge theories. Nevertheless this formalism is of some interest, because in supergravity theories the analog of the conventional formalism is very cumbersome. As described in Chapter 31, the formalism that has been chiefly used to derive physically interesting results in supergravity theories follows an approach like that of de Wit and Freedman, with supersymmetry transformation rules involving covariant derivatives in place of ordinary derivatives, rather than an approach based on conventional supersymmetry transformations like those of Eq. (27.8.1). It is therefore of some interest to understand the relation between the de Wit–Freedman formalism and the conventional approach in the relatively simple context of $U(1)$ gauge theory, and in particular to explain the origin of the extra term in the transformation rule for $\mathcal{F}$.

In writing supersymmetry transformations (27.8.3) that did not involve the components C , M , N , or ω of the gauge superfield V , de Wit and Freedman were implicitly adopting the Wess–Zumino gauge discussed in Section 27.1. But the choice of Wess–Zumino gauge is not invariant under either the conventional supersymmetry transformations (26.2.11)–(26.2.17) or the extended gauge transformations (27.1.17), so once we adopt this gauge both symmetries are lost. We can, however, define a *combined* transformation, acting on fields in Wess–Zumino gauge, which consists of a conventional supersymmetry transformation followed by an extended gauge transformation that takes us back to Wess–Zumino gauge. *This is the de Wit–Freedman transformation* $\tilde{\delta}$ ⁶.

To construct the de Wit–Freedman transformations in this way, note that for a gauge superfield that satisfies the Wess–Zumino gauge conditions $C = M = N = \omega = 0$, the transformation rules (26.2.11)–(26.2.14) give

$$\delta C = 0, \quad \delta \omega = \not{V} \alpha, \quad \delta M = -(\bar{\alpha} \lambda), \quad \delta N = i(\bar{\alpha} \gamma_5 \lambda).\tag{27.8.5}$$

According to Eqs. (27.1.17), we can get back to Wess–Zumino gauge by performing an infinitesimal extended gauge transformation (27.1.13):

$$V \longrightarrow V + \frac{i}{2} [\Omega - \Omega^*],\tag{27.8.6}$$

⁶This was not shown explicitly by de Wit and Freedman. But, in fact, although the point of their paper was to emphasize that the details of the transformations (27.8.3) and (27.8.4) could be inferred from the requirement of a closed supersymmetry algebra (also for non-Abelian gauge theories), they remarked that they had actually found these transformations by identifying the fermionic transformations that survive in Wess–Zumino gauge.

where Ω is a left-chiral superfield with components

$$\phi^\Omega = 0, \quad \psi_L^\Omega = -\sqrt{2}\not{V}\alpha_R, \quad \mathcal{F}^\Omega = -(\bar{\alpha}(\mathbf{1} - \gamma_5)\lambda). \quad (27.8.7)$$

According to Eq. (27.1.11), this extended gauge transformation induces on a chiral superfield of charge q the transformation

$$\delta'\Phi = iq\Omega\Phi. \quad (27.8.8)$$

Using the multiplication rules (26.3.27)–(26.3.29), the transformation of the components of Φ is

$$\begin{aligned} \delta'\psi_L &= -i\sqrt{2}q\phi\not{V}\alpha_R, \\ \delta'\mathcal{F} &= -2iq\phi(\bar{\alpha}_L\lambda_R) - i\sqrt{2}q(\bar{\alpha}_L\not{V}\psi_L), \\ \delta'\phi &= 0. \end{aligned} \quad (27.8.9)$$

Adding this to Eq. (27.8.1) and comparing with Eq. (27.8.4) shows that the de Wit–Freedman transformation is indeed the combination of a conventional supersymmetry transformation and the corresponding extended gauge transformation (27.8.8):

$$\tilde{\delta}\Phi = \delta\Phi + \delta'\Phi. \quad (27.8.10)$$